Task 1: German Tank Problem

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Context

Recall the German Tank (or the “Rocket Science") problem from Exercise #7 in the class. You know that the Nazi Army is assigning sequential integer IDs to their tanks. In other words, if they have M tanks, there is a tank with ID i for all $1 \leq i \leq M$. The allied forces capture n tanks and the IDs of these tanks are $1 \leq x_1 \leq x_2 \leq ... \leq x_n$. Our goal is to estimate M, i.e., the number of tanks that the Nazis have, based on the assumption that the tanks were captured uniformly at random from these M tanks.

1 Part A

Let $\hat{M}_{MLE}$ be the maximum likelihood estimator for M. Compute $\hat{M}_{MLE}$.

2 Part B

Let $\hat{M}_{MEAN} = 2(\sum_{i=1}^{n} x_i/n) - 1$ and $\hat{M}_{MVU} = x_n(\frac{n+1}{n}) - 1$ be two additional candidate estimators for M. Theoretically characterize the expected value of each of the three estimators ($\hat{M}_{MLE}$, $\hat{M}_{MEAN}$ $\hat{M}_{MVU}$) for a given M and N. Comment on the bias of each estimator.

3 Part C

Using different values of M and n, simulate different instances of the German Tank Problem, visualize the mean and variance of these three estimators ($\hat{M}_{MLE}$, $\hat{M}_{MEAN}$, $\hat{M}_{MVU}$) as a function of M and n (i.e., fix M to, say 100, 1000, 10K etc., and for each M, plot the mean and variance of the estimators as a function of n - you can compute the mean and variance by repeating the simulation multiple times for each combination of M and n). Discuss the differences between the estimators in terms of their bias and variance.